Gravitational potential energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate height change to near-surface gravitational energy.

SUCCESS CRITERIA - I can...

1. Choose a height reference.
2. Use ΔU=mgΔh with the stated g.
3. Distinguish potential energy from force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “potential energy” and “reference level” to explain your model or result. Sentence starters are optional.

potential energy: energy associated with the configuration of interacting objects

reference level: the chosen position where potential energy is set to zero

weight: the gravitational force on an object

01 Warm-up: bring back an earlier idea

A 3 kg object slows from 7 to 5 m/s.
Find net work and check its sign.

Word help: work = energy transferred by a force along a displacement

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

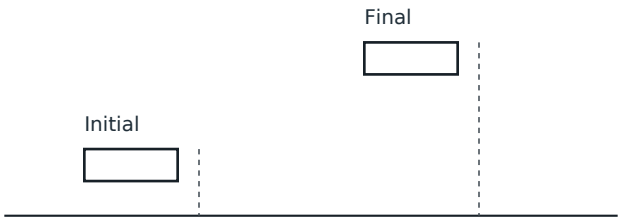
02 Worked example: explain the model

Reference: $\Delta U = mg\,\Delta h = mg(h_f - h_i)$.
Approximately constant g = 10 N/kg. Choose a zero height; the energy change does not depend on that choice.

Worked reasoning: Raising a 2 kg object by 3 m with g=10 N/kg increases gravitational potential energy by mgh=60 J. The chosen zero does not affect this change.

Your turn: Draw initial and final height levels and label the 3 m difference.

Sentence starter: Both ... . They differ in ... because ... .



Choose and label a zero-height level.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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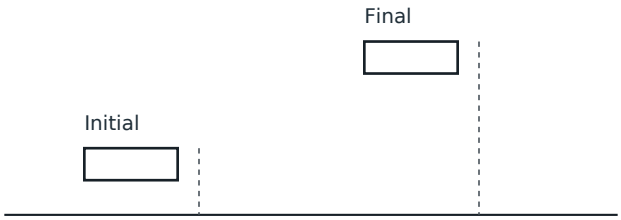
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two identical books sit at heights 1 and 3 m above the floor. Which has more gravitational potential energy relative to the floor?

Sentence starter: I notice ... . This supports ... because ... .



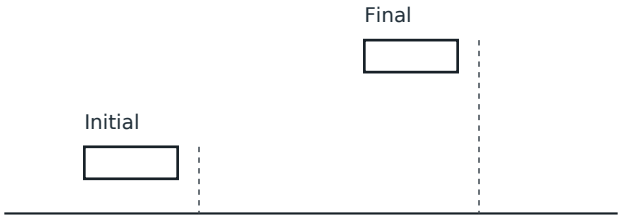
Choose and label a zero-height level.

Marzano check - circle: 1 2 3

04 Apply the physics

A 5 kg load rises from 4 m to 7 m above the floor.
Use g = 10 N/kg.
Find its change in gravitational potential energy. Repeat with U = 0 at a level 2 m above the floor.
Label both height references and explain what stays unchanged.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Choose and label a zero-height level.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Does a negative value of U relative to a chosen zero imply a negative mass or impossible height?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two 4 kg carts at +6 and -4 m/s stick.
Find final velocity and use kinetic energy to explain why this is not elastic.
Use before and after snapshots to organize the momentum account.

Word help: kinetic energy = energy associated with motion

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

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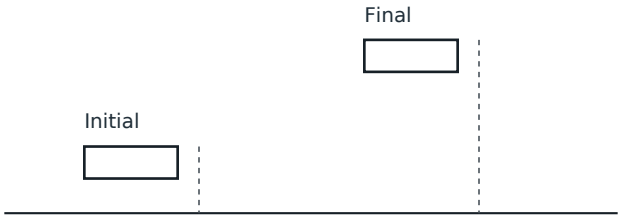
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 2 kg object descends 4 m.
Find ΔU and work by gravity.
Label the relevant heights or distances on the model.
Use g = 10 N/kg when needed.

Sentence starter: I found ... [unit]. This value means ... in this situation.



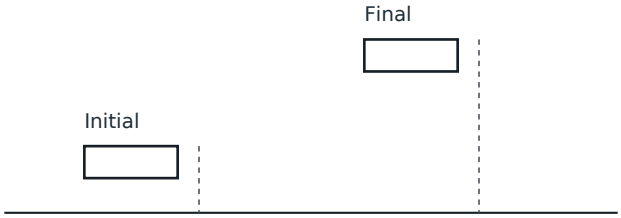
Choose and label a zero-height level.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 3 kg object moves from 2 m below a chosen zero to 5 m above it.
Use g=10.
Find initial U, final U and work by gravity.
Label the relevant heights or distances on the model.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Choose and label a zero-height level.

Marzano check - circle: 1 2 3